AHP Method for Selecting Emergency Care Key Performance Indicators

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Abstract—**The main goal of our study is the ordering of performance criteria and the evaluations of the performance of an Emergency Department (ED). More precisely, this paper deals with the problem of decision-making in the selection of the most important KPIs that have significant contribution when many conflicting criteria exist using the Analytic Hierarchy Process (AHP) method. The application of AHP method in solving multiple criteria decision making is illustrated by our concrete case of the emergency department in Farhat Hached hospital, using Expert Choice (EC) software, which is based on the AHP method. The finally obtained results were analyzed.**

Keywords- ***Key Performance Indicators, Emergency Department, Analytic Hierarchy Process, SMART criteria, Health Care process***

I. INTRODUCTION

Key Performance Indicators (KPIs) allow gathering knowledge and exploring the best way to achieve organization goals. They contribute immensely to the evaluation of process performance. Many researchers have provided different ideas for determining KPIs either manually, and semi-automatically, or automatically which is applied in different fields. However, before measuring KPIs, it is essential that we identify important criteria that should be taken into consideration when developing KPIs for performance monitoring. The aim of using this step is to identify the quality of indicators for comparison and to set priority areas for additional KPI development or improvement. Therefore, organizations deal with many quantitative and qualitative indicators (KPIs) in different areas. The improvement in performance comes from efficient definition and selection of appropriate measurement. Very few organizations really monitor their true KPIs because indicators are mostly quantitative information; it illustrates the processes of a company. Now developing and selecting qualitative KPIs beside the quantitative indicators are very important for planning action plan. Moreover, this selection is generally contextual where the lack of understanding of the performance measures lead to a failure in monitoring and reporting of measures. For instance, indicators useful in department A may be inappropriate for department B. Depending on the context, several kinds of KPIs can be developed, including quantitative or qualitative aspects, and different data source can be used for retrieving and calculating their appropriate values. Choosing the appropriate indicators based on a good understanding of what is important to the organization. In fact, the selection of the right performance measures depends on a number of criteria, including who will use them and what decision they support. In this context, it is recommended that KPIs satisfy the SMART criteria [1, 2, 3] the purpose of AHP method in this research refers to extracting the larger meaning of KPIs and providing a way of thinking when defining key performance indicators. This method is one of the most popular methods for multiple criteria decision making is the Analytic Hierarchy Process (AHP) [4] [5].

Before the KPI are selected, it is necessary to identify and to clarify the criteria which are going to influence the choice of these indicators. The choice makes a difference. If the wrong KPI is measured, or if it is measured in the wrong way, the information may be misleading and the quality of decisions could be affected. Human judgments about values and choices are still the basis for making good decisions. The literature abounds with papers defining the characteristics of metrics and KPIs. Each indicator should be based on criteria that make it suitable for further analysis. Reviewing the literature, it is found that SMART criteria are most often referenced.

In the following sections, we present some previews work. Next, we describe our proposed approach. After that, we identify hospital performance indicators (quantitative-qualitative). In section V, we explore some results based on the integration of Analytical Hierarchy Process (AHP) technique, and SMART criteria (Specific, Measurable, Attainable, Realistic and Time-based). Finally, a brief summary is given in the conclusions section.

II. RELATED WORKS

According to Shahin and al [6], if applying the AHP-SMART method, KPIs can both be prioritized and evaluated in which are more "SMART" than others. In [6] they suggest an integrated approach to the AHP and SMART criteria for prioritizing KPIs where a case study was conducted in a hotel area. Despite this, according to the authors, literature reviews show that there have been limited efforts in designing a standardized method for KPI prioritization.

In [12] ORTEGA affirms that in order to define good PPIs, it is recommended that they satisfy the SMART criteria. In her thesis, she describes the template for the definition of PPIs. This template has been defined in order to fulfill the SMART criteria. However, all indicators are previously supposed SMART and she just fulfilled in a very simple way all necessary fields in the template. For example, for the field Goals connecting to the Relevant characteristic, there is no guarantee that this KPI contributes really to evaluate the process performance. So the relevance of KPIs should be evaluated from the expert point of view. In our work, we cover this shortcoming when using AHP and starting with constructing an overall goal which is selecting and ranking KPIs according to the BP objective and then specifying the components that have a negative or positive impact on reaching the goal. In [13], the authors elaborate a relatively small number of key performance indicators (KPIs) for measuring Occupational safety and health management systems (OSH MSs) operational performance. For the selection of KPIs, the AHP method was employed. This study includes 109 indicators grouped in 20 subsets corresponding to 20 main components of the adopted model of OSH management system. The ranking and prioritization of leading performance indicators were made in relation to a set of SMART criteria. The highest significance has been attributed to the „„Relevant"" criterion. Therefore, the goal of the performed prioritization of PPIs was not the selection of indicators being just relevant to the measurement of OSH MS performance, but the selection of indicators being the most relevant, and thus the reduction in measurement and monitoring actions down to a much lower number of KPI. However, in some cases, we can detect indicators who have a higher score with respect to the relevance criterion and have a lower score in other criteria. In the Final phase, when AHP combines the criteria weights and the KPIs scores, thus determining a global score for each KPI, and a consequent ranking. The global score for a given KPI is a weighted sum of the scores obtained with respect to all the criteria. As a consequence, this relevant indicator can have "bad" ranking in the final phase. For this reason, in our work, we aim to improve the definition of KPIs by proposing a new approach in which we guarantee that all indicators are specific and measurable and just the other criteria make difference in calculating priorities. In [14], the authors investigate the use of AHP in estimating determinants of patient satisfaction towards service quality delivery of public teaching hospitals in Southwest Nigeria. They use AHP-based approach to measure the quality of service (e.g. Tangibility, Reliability, Responsiveness, Assurance, Empathy, Effective communication and Waiting time) rendered by the hospitals from the patient"s perspectives. The questionnaire used in this paper revealed that the majority of respondents (patients) were female and out of the 420 copies of a questionnaire distributed to the patients, 348 copies of the questionnaire were returned showing 82.9 % response rate and only 326 copies of the questionnaire were found useful for the AHP analysis. As a consequence, if we change the Socio-demographic component like age, gender, marital status, and education, or change the medical and paramedical staff availability or team, all this aspect related to the relationship between patients" expectations and experiences of the received treatment will also change. Moreover, there was a significant association between patient satisfaction (e.g. for cleanliness of the hospital, up-to-date medical equipment and for waiting time alternatives) and the period of investigation (record patient satisfaction during the afternoon, night or weekend). This can have an influence on the determination of the priority weight for service quality dimensions and ranking of KPIs. We can conclude that the variation in the views of the people participating in the judgmental process might lead to different results. To be more effective and to guarantee that the ranking criteria and alternative are up to date, in our work we propose a new approach described in the following section.

III. Proposed approach

In general, indicators are constantly under pressure to control rapidly the business process. As a result, it"s crucial to continue studying the efficiency of existing or proposed indicators and exploring improvement opportunities. Lord Kelvin defined KPIs as "When you can measure what is spoken about and measure it in numbers, you know something about it, when you cannot express it in numbers, your knowledge is of meager and unsatisfactory kind; it may be the beginning of knowledge but you have scarcely, in your thoughts advanced to the stage of science"[15].The development of these indicators based on the SMART criteria is crucial prior to their actual implementation.

In the first phase of the suggested approach, we essentially deal with various measurement related to the business processes. We will also deal with two different aspects: qualitative and quantitative. In each aspect we define and validate the appropriate indicators; these measures should have a significant impact on the organization. In the second and third phase, we identify all the different criteria that affect the decision and organize all indicators using AHP method. In the last phase, we check if the results of the application of AHP are compatible with the expectations, a review of the previous results is needed. It is very important to avoid gaps between the KPIs definition and KPIs priority. Whenever necessary, the first list of indicators needs to be complemented to include other indicators not previously identified or considered. Also, it is important to know all the reasons that supported how and why an indicator has a good weight in one criterion than another one. This reviewing can be helpful to improve some indicators definition or judgment respecting the appropriate criteria. The continuous life cycle guarantee that KPI priorities are up to date and allowing a continuous improvement of the decision-making process.

Some advantages of the proposed approach are as follows: (1) Evaluation of SMART measurement taking into account both quantitative and qualitative aspects; (2) Giving in the last phase the possibility to involve all informed persons in establishing additional indicators, thus improving the definition of other indicators or concentrating on the most valuable indicators based on their priorities; (3) Obtaining a primary list of the right "SMART" performance measures before the execution in the business process management system.

IV. KEY PERFORMANCE INDICATORS

According to Parmenter [7], key performance indicators (KPI) represent a set of measures focusing on those aspects of organizational performance that are the most critical for the current and future success of the organization. A process performance measure is a quantity that can be unambiguously determined for a given business process assuming of course that the data to calculate this performance measure is available [8]. In the illustrated case study, after discussion with experts, we arranged a list of relevant indicators. From this analysis, we took into account 15 quantitative indicators And 19 qualitative indicators related to the patient instance.

Activities	*Quantitative indicators*	
	duration	*waiting time*
Registration	Quanti_KPI7 The duration of registration by patient	Quanti_KPI17 The waiting time before registration and payment
Sorting	Quanti_KPI8 The duration of sorting by patient	Quanti_KPI18 The waiting time before sorting
Consultation	Quanti_KPI9 The duration of consultation by patient	Quanti_KPI19 The waiting time before consultation in box
Delayed emergency	Quanti_KPI11 The duration of delayed emergency by patient	Quanti_KPI22 The waiting time before consultation in delayed emergency
Lying waiting	Quanti_KPI10 The duration in supervision room by patient	Quanti_KPI20 The waiting time before consultation in the supervision room
Shock treatment	Quanti_KPI12 The duration in crash room by patient	Quanti_KPI21 The waiting time before consultation in crash room

TABLE I. LIST OF QUANTITATIVE INDICATORS RELATED TO BUSSINESS PROCESS TASK

Table 1 shows all KPIs for each activity. For example, in the column labeled duration, we found all necessary measurement related to the duration of the every task. The duration is calculated from the started task to the completed registration task event transition. In the column labeled waiting duration, the KPIs related to how long a patient waits before the consultation. We add 3 aggregated quantitative indicators related to patient (instance) such as the sum of all previous activities duration, the sum of waiting time per all patients in all activities and the process duration by the patient includes the waiting time and the real duration of all activities. In our work, we concentrate on indicators related to patient process instances because if we manage and well define the "base measure" consequently the aggregated indicators will be managed because they provide a global overview of the performance of the process.

By addressing patient queries, the qualitative aspect will be in a better position to improve satisfaction toward the healthcare processes. Those indicators concern the qualitative aspects of care in the ED, such as staff attitudes towards patients, but also, they concern some quantitative aspect such as para medical staff availability and the overall waiting time before treatment by paramedical personnel. This is due to the fact that sometimes the nurse is available for example but the patients who are in less urgent are not given sufficient attention by nursing staff. So, in order to provide a higher level of quality and consistency with quantitative measurement and with the processes in place, we record the patient"s level of satisfaction with those indicators. The aim of this questionnaire is to determine why patients are unsatisfied and what we can do to make it better.

For the improvement of key performance indicator in one hand, we use all the measurement to prioritize KPIs based on an integration of analytical hierarchy process and SMART criteria. On the other hand, we are interested especially in the quantitative that are related process indicators and the qualitative indicators according to patient satisfaction level. Doing so would make the performance indicators relevant and results useful.

V. INTEGRATION OF AHP AND SMART KPIS

This section, we identify healthcare performance indicators and validate these indicators list by the expert. We present AHP technique and we describe the selected KPIs according to their degree of importance. Indicator items were examined by SMART criteria. The data was analyzed by Expert Choice software.

A. Analytic Hierarchy Process method

The AHP method is a basic approach to decision making. AHP is referred to as the most powerful and widely used technique for decision making. It allows decision maker(s) to measure the consistency and stability of their decisions [3]. It is designed to develop overall priorities for ranking the alternatives with respect to several criteria.

The AHP approach was chosen since it was essential to collect data from experts who were highly experienced in healthcare process and thereby make a positive contribution to the identification of optimum definition for key performance measures. Also, the few number of criteria (SMART) seemed appropriate in this paper to structure the decision maker"s mind in order to provide a systematic prioritization of sustainability performance indicators.

The AHP both allows for inconsistency in the judgments and provides a means to improve consistency [9]. To make comparisons, we need a scale of numbers that indicates how many times more important or dominant one element is over another element with respect to the criteria or property with respect to which they are compared. The scale ranges from "equal" (number 1) to "absolutely more important than" (number 9) [10].

The meaning of the table is that criterion A is strongly more important than criterion B. To obtain better insight of such KPI and to identify the most significant criteria AHP method proposes to create as many refinements as needed for the specific problem and to estimate verbally the value of each new point of the scale. The application of the AHP method is based on the following three steps:

- Defining the problem and creating a hierarchical model of decision problems: This structure consists of a goal on the top level, criteria, and sub-criteria through intermediate levels and alternatives to the lowest level.
- Comparing the relative importance among the elements of the hierarchy by making a series of judgments: AHP uses simple pairwise comparisons based on the Saaty scale. AHP has three judgment elicitation modes (verbal, numerical, or graphical) by which a decision maker can provide judgments about the relative importance of objectives or criteria. After the judgments are entered, we check whether the input data satisfies a consistency test. The inconsistency ratio should be less than 0.1 or so to be considered reasonably consistent. If it does not, we redo the pairwise comparisons.
- Establishing priorities (i.e., weights) for each node of the hierarchy by calculating the relative weights of the decision elements and then aggregating the relative weights to obtain scores and hence rankings for the decision alternatives. These steps are accomplished by using software such as Expert Choice. Expert Choice (EC) is a package for decision support, based on the analytic hierarchy process.

B. *SMART criteria*

There is no clear consensus about what the five SMART keywords mean. Many authors use the "SMART" Rule as a means of identifying the characteristics [11]:

- Specific: The KPI is clear and focused toward performance targets or a business purpose.
- Measurable: The KPI can be expressed quantitatively.
- Attainable: The targets are reasonable and achievable.
- Realistic or relevant: The KPI is directly pertinent to the work done on the project.
- Time-Based: The KPI is measurable within a given period of time".

The SMART rule was originally developed for establishing meaningful objectives for projects and later adapted to the identification of metrics and KPIs [11]. Applying the AHP methodology into a KPI means evaluating alternatives with respect to previously determined criteria. It's necessary to consider all relevant criteria and make judgments according to professional experience so that results give us adequate answers. Having KPIs defined in natural language, it is harder to fulfill this SMART criteria. For this reason, typical questions that may guide the selection of indicators for each criterion in our use case are listed below:

- Is the indicator clearly explained? What it measures?, How it measure? And when it measure?
- Is the indicator measurable? Is it quantifiable using an available tool?
- What is the most important indicator to meet its target value?
- What is the most important control indicator taking into account the resources and the means available?
- What is the most important indicator to consider in a well defined period of time?

During the rest of this section, we present a practical application of AHP method.

C. *AHP hierarchy*

In our case, there are two hierarchical decision models, one for the quantitative indicators and the other for qualitative indicators. The AHP tree is structured into different levels as follows: (1) Level 1: the overall goal is the ranking and prioritization of performance indicators. (2) Level 2: SMART criteria and (3) Level 3: list of KPI. In order to class this list in descending order of relevance, the research conducts a survey involving experts who are directly involved in health care process activities.

D. *pairwise comparison matrix*

In this step, we start by setting criteria priorities. The first criteria pairwise comparison matrix will be used to establish the relative importance of five main criteria. In this step, the decision maker needs to define the most important criteria for quantitative indicators and for qualitative indicators. The pairwise comparison for the criteria under the Goal node is given in Figures 1 and 2.

	Specific	Mesurable	Attainable	Realistic	Time based
Specific		1,0	2,0	3,0	2,0
Mesurable			2,0	3,0	2,0
Attainable				3,0	2,0
Realistic					3,0
Time based	Incon: 0,03				

Figure 1. Pairwise comparison for qualitative indicators

	Specific	Mesurable	Attainable	Realistic	Time-Based
Specific		3,0	2,0	1,0	1,0
Mesurable			2,0	2,0	1,0
Attainable				2,0	1,0
Realistic					2,0
Time-Based	Incon: 0,03				

Figure 2. Pairwise comparison for quantitative indicators

Figures 1 and 2 show that all the judgments are considered consistent and accepted for analysis (Incon:

0.03). Secondly, all the indicators have been compared, by grouping them into pairs with respect to their SMART criteria. In this case, we have to find first the most important KPIs from the list of KPI candidates. The involved experts answered some questions particularly according to each criterion when alternatives are being compared with each other in pairs. These experts are more aware of the problems that KPIs have to cope with. For instance, some examples for each decision alternative are illustrated below.

TABLE II. COMPARISONS FOR THE QUANTITATIVE ALTERNATIVES UNDER « ATTAINABLE » AND « REALISTIC » CRITERIA

Alternatives under « Attainable » and « Realistic » criteria	
Alternatives and judgment	***Explication***
All indicators related to the sorting, consultation, Lying waiting and Shock treatment are also moderately more important than other indicators (value 3 according to figure 1)	indicators belonging to the category patients who need simple consultation or surgical consultation or not stable patients who receive care and treatment or patient in serious harm who require immediate medical care. Also, patients in sorting require a ranking during sorting to determine the priorities for support. are also moderately less important than other indicators
All indicators related to the registration and delayed emergency task	Patients in this category are in general stable and not urgent patients that cannot be in charge quickly by the medical staff. And patient before registration represents patients prior to first medical contact

Similarly, we present the results of judgments for the qualitative alternative. Some results are summarized in pairwise comparison judgment matrices as shown in figure 3 and 4.

	KPI1	KPI2	KPI3	KPI4	KPI5	KPI6	KPI7	KPI8	KPI9	KPI10	KPI11	KPI12	KPI13	KPI14	KPI15	KPI16	KPI17	KPI18	KPI19
KPI1		1,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI2			2,0	2,0	2,0	1,0	1,0	1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI3				1,0	1,0	2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
KPI4					1,0	2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
KPI5						2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
KPI6							2,0	1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI7								1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI8									1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI9										2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI10											1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
KPI11												1,0	1,0	1,0	1,0	1,0	1,0	1,0	1,0
KPI12													1,0	1,0	1,0	1,0	1,0	1,0	1,0
KPI13														1,0	1,0	1,0	1,0	1,0	1,0
KPI14															1,0	1,0	1,0	1,0	1,0
KPI15																1,0	1,0	1,0	1,0
KPI16																	1,0	1,0	1,0
KPI17																		1,0	1,0
KPI18																			1,0
KPI19																			

Figure 3. Pairwise comparison of qualitative alternatives with respect to Measurable

In figure 3, KPI1, KPI2, KPI6, KPI9, KPI7, KPI8 are not measurable because they are rather related to the behavior, attitudes, and practices of the staff. However, KPI3, KPI4, KPI5, KPI10, KPI11, KPI12, KPI13, KPI14, KPI15, KPI16, KPI17, KPI18, KPI19 can be quantified so these indicators are also moderately more important compared to others.

	KPI1	KPI2	KPI3	KPI4	KPI5	KPI6	KPI7	KPI8	KPI9	KPI10	KPI11	KPI12	KPI13	KPI14	KPI15	KPI16	KPI17	KPI18	KPI19
KPI1		1,0	2,0	2,0	2,0	1,0	2,0	2,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0
KPI2			2,0	2,0	2,0	1,0	2,0	2,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0
KPI3				1,0	1,0	2,0	1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI4					1,0	2,0	1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI5						2,0	1,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI6							2,0	2,0	1,0	2,0	2,0	2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0
KPI7								1,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI8									2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0	2,0
KPI9										2,0	2,0	2,0	2,0	2,0	2,0	1,0	1,0	1,0	1,0
KPI10											1,0	1,0	1,0	1,0	1,0	2,0	2,0	2,0	2,0
KPI11												1,0	1,0	1,0	1,0	2,0	2,0	2,0	2,0
KPI12													1,0	1,0	1,0	2,0	2,0	2,0	2,0
KPI13														1,0	1,0	2,0	2,0	2,0	2,0
KPI14															1,0	2,0	2,0	2,0	2,0
KPI15																2,0	2,0	2,0	2,0
KPI16																	1,0	1,0	1,0
KPI17																		1,0	1,0
KPI18																			1,0
KPI19																			

Figure 4. Pairwise comparison of qualitative alternatives with respect to Attainable

In figure 4, Indicators related to human and material resources (the number of doctors and nurses, the number of beds and medical equipment) are also moderately more important than other indicators since the values of these indicators always have to be controlled to provide a fast efficient care for patients.

E. Synthesis

After all, responses were recorded; the priorities of alternatives with respect to each criterion are automatically computed. Expert Choice tool synthesizes the priorities of the objectives, data, and ratings to determine the overall priorities of the alternatives. These priorities are a corporate perception of what is important. The AHP synthesis for this use case is given in figure 5.

KPI12 The duration in crash room by patient	,087
KPI21 The waiting time before consultation in crash room	,084
KPI20 The waiting time before consultation in the supervision room	,074
KPI8 The duration of sorting by patient	,072
KPI18 The waiting time before sorting	,070
KPI19 The waiting time before Consultation in box	,070
KPI9 The duration of Consultation by patient	,069
KPI10 The duration in supervision room by patient	,069
KPI7 The duration of registration by patient	,066
KPI11 The duration of delayed emergency by patient	,057

Figure 5. Synthesis of top 10 quantitative indicators with respect to the goal

In figure 5, the most important indicators are the duration in the crash room, followed by the waiting time before consultation in the crash room and the waiting time before consultation in the supervision room.

KPI3 paramedical staff availability	,065
KPI4 medical staff availability	,065
KPI5 medical equipment availability	,065
KPI18 The regularity doctor visits in the supervision room	,065
KPI19 The regularity doctor visits in the crash room	,065
KPI8 the quality of care for patients by medical staff	,062
KPI7 the quality of care for patients by paramedical staff	,061
KPI16 The overall waiting time before treatment by paramedical personnel	,053
KPI17 The overall waiting time before treatment by medical personnell	,053
KPI10 installation in the waiting room before UD (comfort, cleanliness, hygiene...)	,046

Figure 6. Synthesis of top 10 qualitative indicators with respect to the goal

Figure 6 shows the priorities of the KPI with respect to the overall goal. For selection, the qualitative indicators are now arranged in overall priority. The availability of medical staff and medical equipment is the highest rated indicator, the regular visit is the next highest rated indicator and the quality of care is the third group rated indicators.

We can also calculate values of each alternative for each criterion. In Figure 7, qualitative indicators under "specific" criteria are sorted by priority. We can see for example that the indicator related to the clarity of information is not specific enough. So, in the proposed approach, we can improve the definition of this indicator by indicating in which condition, which categories of involved patient, and with which type of consultation. For another example, that of qualitative_KPI1, there is no indication of the activity (where and when) in which interest and attention is brought by paramedical staff.

Priorities with respect to:

prioritization of SMART performance indicators
>Specific

KPI1 interest and attention brought by paramedical staff	,034
KPI2 interest and attention brought by medical staff	,034
KPI3 paramedical staff availability	,034
KPI4 medical staff availability	,034
KPI5 medical equipment availability	,034
KPI6 hospital staff is well dressed	,034
KPI7 the quality of care for patients by paramedical staff	,034
KPI8 the quality of care for patients by medical staff	,034
KPI9 The clarity of information	,034
KPI10 installation in the waiting room before UD (comfort, cleanliness, hygiene...)	,069
KPI11 installation in the waiting room before registration	,069
KPI12 installation in the waiting room before consultation in box room	,069
KPI13 installation in the sorting room	,069
KPI14 installation in the supervision room	,069
KPI15 installation in the crash room	,069
KPI16 The overall waiting time before treatment by paramedical personnel	,069
KPI17 The overall waiting time before treatment by medical personnell	,069
KPI18 The regularity doctor visits in the supervision room	,069
KPI19 The regularity doctor visits in the crash room	,069

0 missing judgments.

Figure 7. Priorities of qualitative indicators with respect to the goal under specific criteria

Up to this point, it should be noted that AHP method results depending on the context where some indicators may require reformulation or new indicators may be needed to take into consideration the specificity of the context in which they are applied. Our motivation using the AHP method is the limited number of the criteria and the evaluation of the consistency ratio of judgments. If the ratio unacceptable, we can revise pairwise comparisons. The best thing to do after analyzing AHP synthesis result is going back and revise the KPIs measurement in order to get "SMART" KPIs.

VI. CONCLUSION

Analytic Hierarchy Process is a very useful technique for multi-criteria decision-making. It enables people to make decisions involving many kinds of concerns including planning, setting priorities, selecting the best among a number of alternatives. In this paper, an approach is proposed using AHP to prioritize key performance measures of health care organizations. Therefore an application of AHP to a decision-making problem from health care field is described in this paper. As part of our future work, we plan to extend our approach by integrating our approach in the Business process management life cycle.

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